

BI Strategy Note & Operations Portfolio

SECTION 1: CONTEXT

This strategy note outlines the deployment of an AI-enabled Business Intelligence framework for HimShakti's bio-waste processing division. The primary decision-maker is the Chief Operations Officer (COO), who requires real-time transparency into supply chain yields, processing plant efficiencies, and financial leakages to optimize operational margins.

SECTION 2: KPIs RECOMMENDED

- **Inbound Weight Capacity (KG):** Total volume of raw agricultural biomass received at processing facilities.
- **Processing Waste Mass (KG):** Total volume of organic discard generated during the extraction phase.
- **Net Biomass Yield (%):** The percentage efficiency calculated by dividing usable output mass by total inbound raw mass.
- **Aggregate Financial Loss (INR):** The total calculated monetary value of wasted crop mass per batch.
- **Processing Center Efficiency Score:** A comparative metric ranking plant operational productivity based on waste minimized per ton.
- **Fulfillment Cycle Time (Hours):** Total elapsed time from raw crop arrival at a center to finalized extract packaging.

SECTION 3: DATA SOURCES

The dashboard aggregates data from three primary operational streams. First, digital weighbridge logs capture real-time inbound raw crop weights upon arrival at centers. Second, shop-floor IoT sensors track batch extraction volumes and organic residue weights during processing. Third, enterprise resource planning (ERP) databases link market prices to waste mass, calculating financial losses. These pipelines merge into a centralized cloud store for analysis.

SECTION 4: TOOL STACK RECOMMENDED

We utilize a cost-effective, free-tier modern data stack. Data ingestion is orchestrated via Python scripts running on local servers. Storage and basic warehousing utilize the Google Sheets Cloud API as an accessible relational database. For core analytics and visualization, Looker Studio is deployed for building interactive corporate dashboards. Finally, Gemini AI is leveraged for automated anomaly detection and strategic reporting narratives.

SECTION 5: PHASED IMPLEMENTATION PLAN

Deployment will execute across three immediate phases. **Phase 1 (Weeks 1–2):** Establish data ingestion pipelines by connecting weighbridge logs and inventory sheets to the cloud repository. **Phase 2 (Weeks 3–4):** Build and validate core Looker Studio metric scorecards and interactive trend charts. **Phase 3 (Week 5+):** Embed Gemini AI prompt workflows to automate predictive loss alerts and generate monthly operational reviews.

SECTION 6: BI CASE STUDIES SUMMARY

Case Study 1: Swiggy (Logistics Optimization)

- **Stack:** Python, Spark, AWS Redshift, Snowflake, Tableau
- **Source:** <https://bytes.swiggy.com>
- **Problem & Scope:** Optimizing real-time delivery routing and balancing driver supply with hyper-local order spikes during peak hours. Tracked KPIs included Cost Per Delivery (CPD), Order-to-Delivery Time, Driver Utilization Rate, and Order Cancellation Rate utilizing GPS streams and transactional order histories.
- **Outcome:** Swiggy successfully lowered driver idle times and improved ETA accuracy by over 15%, reducing delivery overheads.

Case Study 2: Flipkart (Supply Chain Forecasting)

- **Stack:** Hadoop, Apache Hive, Presto, Python, Power BI
- **Source:** <https://tech.flipkart.com>
- **Problem & Scope:** Managing inventory stockouts and preventing warehouse overstocking across regional distribution hubs before major seasonal sales events. Tracked metrics included Inventory Turnover Ratio, Days Inventory Outstanding (DIO), Stockout Rate, and Fulfillment Lead Time driven by customer clickstreams and logistics timelines.
- **Outcome:** Enabled automated, demand-driven inventory placement, decreasing nationwide carrying costs by roughly 12%.

SECTION 7: BI MATURITY MODEL MAPPING

Dimension	Emerging	Developing	Established	Leading
Data Availability		X (HimShakti)		
Tooling		X (HimShakti)		
Team		X (HimShakti)		
Use of Insights		X (HimShakti)		

Maturity Justification: HimShakti is firmly positioned in the Developing stage. While the organization possesses foundational operational data streams, historical datasets are siloed across disconnected centers. The current transition to free-tier cloud BI tools like Looker Studio marks a shift away from manual Excel dependencies. Insights are currently actioned periodically rather than driving real-time strategy, which will change as dedicated analytics responsibilities